



**Sydney Girls High School
2025
Trial Higher School Certificate
Examination**

Mathematics Extension 2

**General
Instructions**

- Reading time – 10 minutes
- Working time – 3 hours
- Write using a black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

**Total marks:
100**

Section I – 10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 90 marks

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

Name: Teacher: 	THIS IS A TRIAL PAPER ONLY It does not necessarily reflect the format or the content of the 2025 HSC Examination Paper in this subject.
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Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

Questions	Marks
1 What is the converse of the statement “if you can’t see my mirrors, then I can’t see you”?	1
A. If I can’t see your mirrors, then you can’t see me.	
B. If I can’t see you, then you can’t see my mirrors.	
C. If you can see me, then I can see your mirrors.	
D. If I can see you, then you can see my mirrors.	
2 What is the contrapositive of the statement “if you can’t see my mirrors, then I can’t see you”?	1
A. If I can’t see your mirrors, then you can’t see me.	
B. If I can’t see you, then you can’t see my mirrors.	
C. If you can see me, then I can see your mirrors.	
D. If I can see you, then you can see my mirrors.	

- 3 If $\underline{u} = 5\underline{i} + (t - 3)\underline{j} + 3\underline{k}$ and $\underline{v} = \underline{i} + (t + 9)\underline{j} + 2\underline{k}$ are perpendicular, what are the possible values of t ? 1

- A. $t = 8, 2$
- B. $t = -8, -2$
- C. $t = 8, -2$
- D. $t = -8, 2$

- 4 Suppose $\underline{a}$, $\underline{b}$ and $\underline{c}$ are three vectors such that $\underline{a} + \underline{b} + \underline{c} = \underline{0}$, $|\underline{a}| = 3$, $|\underline{b}| = 5$, and $|\underline{c}| = 7$. 1

What is the angle between $\underline{a}$ and $\underline{b}$?

- A. $\frac{\pi}{6}$
- B. $\frac{2\pi}{3}$
- C. $\frac{5\pi}{3}$
- D. $\frac{\pi}{3}$

- 5 A particle of mass m moves so that its velocity v is given by $v = f(x)$. 1

The resultant force that causes this motion is given by:

- A. $mf'(x)$
- B. $mx f'(x)$
- C. $mf'\left(\frac{1}{2}x^2\right)$
- D. $mf(x)f'(x)$

- 6** Let ω be a non-real cube root of unity. **1**

What is the value of $(1 + 2\omega + 3\omega^2)(1 + 3\omega + 2\omega^2)$?

- A. 3
- B. 6
- C. 9
- D. 12

- 7** Suppose that z_1 and z_2 are two distinct non-zero complex numbers satisfying $|z_1 + z_2| = |z_1 - z_2|$. **1**

Which of the following is always true?

- A. $z_1 z_2$ is purely imaginary.
- B. $\frac{z_1}{z_2}$ is purely imaginary.
- C. $\frac{z_1 + z_2}{z_1 - z_2}$ is purely imaginary.
- D. None of the above.

- 8 A particle moves from the initial conditions $t = t_0$, $x = x_0$, $v = v_0$ to the final conditions $t = t_F$, $x = x_F$, $v = v_F$ under a force that produces an acceleration of $\ddot{x}$. 1

Which of the following is incorrect?

- A. If $\ddot{x} = f(t)$, then $v_F = \int_{t_0}^{t_F} f(t) dt + v_0$.
- B. If $\ddot{x} = f(v)$, then $x_F = \int_{v_0}^{v_F} \frac{f(v)}{v} dv + x_0$.
- C. If $\ddot{x} = f(v)$, then $t_F = \int_{v_0}^{v_F} \frac{1}{f(v)} dv + t_0$.
- D. If $\ddot{x} = f(x)$, then $(v_F)^2 = 2 \int_{x_0}^{x_F} f(x) dx + (v_0)^2$.

- 9 Suppose that a , b and c are non-zero real numbers satisfying 1

$$\frac{a^2}{b^2 + c^2} < \frac{b^2}{c^2 + a^2} < \frac{c^2}{a^2 + b^2}.$$

Which of the following statements can be deduced from the preceding inequality?

- A. $a < b < c$
- B. $|a| < |b| < |c|$
- C. $c < b < a$
- D. $|c| < |b| < |a|$

- 10** Three non-zero vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ on a number plane are such that $\underline{b} \cdot \underline{c} = 2|\underline{a}|$, $\underline{a} \cdot \underline{c} = 3|\underline{b}|$, and $\underline{a}$ is perpendicular to $\underline{b}$. **1**

What is the minimum value of $|\underline{c}|$?

- A. $2\sqrt{2}$
- B. $2\sqrt{3}$
- C. $2\sqrt{5}$
- D. None of the above.

Examination continues overleaf...

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question on the writing paper supplied. Start each question on a NEW page. Extra writing paper are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on a NEW page		Marks
(a)	Give a counterexample to disprove the following statement: $\text{For all } x, y \in \mathbb{R}, \text{ if } x^2 \geq y^2, \text{ then } x \geq y.$	1
(b)	Write down the negation of the following statement: $\text{There exists } k \in \mathbb{R} \text{ such that } k^2 < 0.$	1
(c)	Consider the vectors $\underline{u} = 3\underline{i} + 2\underline{j} + \underline{k}$, $\underline{v} = -5\underline{j} + 2\underline{k}$ and $\underline{w} = 2\underline{i} + 2\underline{j} + 5\underline{k}$.	
	(i) Prove that $\underline{v}$ and $\underline{w}$ are perpendicular.	1
	(ii) Verify that $ \underline{u} + \underline{v} + \underline{w} \leq \underline{u} + \underline{v} + \underline{w} $.	2
(d)	Given the points $A(1, 3, 5)$, $B(2, -4, 11)$ and $C(-3, -1, 0)$, find $\angle ABC$. Give your answer correct to the nearest minute.	3
(e)	By using integration by parts, evaluate $\int_0^{\frac{\pi}{2}} x \cos x \, dx$.	3
(f)	Evaluate $\text{Arg}(i^{2025})$, where $\text{Arg}(z)$ denotes the principal argument of z .	2
(g)	Prove that $\sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta})$.	2

End of Question 11

Question 12 (15 marks) Start on a NEW page**Marks**

(a) Find $\int \cos^3 x \sin^2 x \, dx$. **2**

(b) By using an appropriate t -substitution, or otherwise, find $\int \frac{d\theta}{3 + 2 \cos 4\theta}$. **3**

(c) (i) By expanding $(\cos \theta + i \sin \theta)^3$ and using de Moivre's theorem, show that **2**

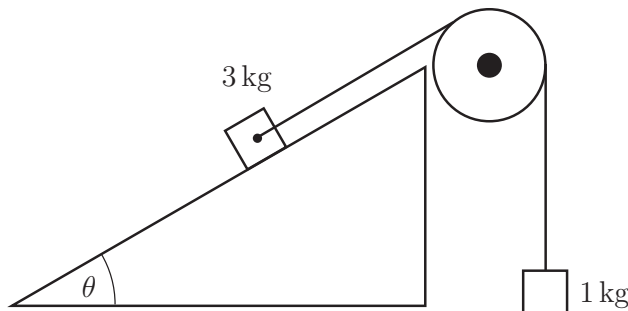
$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

(ii) Hence, or otherwise, solve $8x^3 - 6x + 1 = 0$ where $x = \cos \theta$. **3**

(iii) Deduce that **2**

$$\cos \frac{\pi}{9} - \cos \frac{2\pi}{9} - \cos \frac{4\pi}{9} = 0.$$

(d) The diagram below shows particles of mass 1 kg and 3 kg connected by a light inextensible string passing over a smooth pulley. The 3 kg mass lies on a smooth plane inclined at an angle of θ to the horizontal, and T is the tension in the string. **3**



Find θ so that the system will be in equilibrium. Give your answer in degrees, correct to one decimal place.

End of Question 12

Question 13 (15 marks) Start on a NEW page**Marks**

- (a) The velocity v of a particle moving along the x -axis is given by

$$v^2 = 12 + 8x - 4x^2,$$

where x is the displacement of the particle from the origin O .

- (i) Prove that the particle is undergoing simple harmonic motion. **2**
- (ii) Find the period, amplitude and centre of motion. **2**
- (b) Prove by contradiction that $\sqrt[3]{5}$ is irrational. **2**
- (c) Sketch the region of the complex plane defined by $\operatorname{Re}(z^2) \leq (\operatorname{Re} z)^2$ and $-\frac{2\pi}{3} \leq \operatorname{Arg}(z + i) \leq 0$. **3**
- (d) By using the substitution $u^6 = x$, or otherwise, evaluate $\int_1^{64} \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx$. **3**
- (e) Find $\int \frac{x^2 + 1}{x^4 + 8x^2 + 12} dx$. **3**

End of Question 13

Question 14 (15 marks) Start on a NEW page**Marks**

- (a) Two spheres $\mathcal{S}_1$ and $\mathcal{S}_2$ have Cartesian equations $x^2 + y^2 + z^2 = 9$ and $(x - 3)^2 + (y + \sqrt{6})^2 + (z - 7)^2 = 25$ respectively.
- (i) Prove that $\mathcal{S}_1$ and $\mathcal{S}_2$ are externally tangent. **3**
- (ii) Find the vector equation of ℓ_1 , where ℓ_1 is the line through the centres of $\mathcal{S}_1$ and $\mathcal{S}_2$. Give your answer in the form $\underline{r} = \lambda \underline{d}$. **2**
- (iii) Consider the line ℓ_2 with vector equation **2**

$$\underline{r} = \begin{pmatrix} 1 \\ -\sqrt{6} \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}.$$

Determine any points of intersection between lines ℓ_1 and ℓ_2 .

- (iv) Deduce, with mathematical justification, whether the lines ℓ_1 and ℓ_2 are skew. **2**
- (b) The rise and fall of the tide at a certain port may be considered as simple harmonic, with the time difference between successive high tides being 10 hours. The harbour entrance has a depth of 20 m at high tide and 8 m at low tide. **3**
- If low tide occurs at 10 a.m. on a certain day, find the earliest time that a cargo ship requiring a minimum depth of 15 m of water can pass through the entrance.
- (c) Let a and b be consecutive positive integers with $a < b$. **3**
- By considering the expansion of $(a^2 + a + 1)^2$, prove that if $c = ab$, then $\sqrt{a^2 + b^2 + c^2}$ is an odd integer.

End of Question 14

Question 15 (15 marks) Start on a NEW page

Marks

- (a) Let z and w be two distinct non-zero complex numbers. **3**

Show that if $|z|^2 = |w|^2 = 2 \operatorname{Re}(z\bar{w})$, then the triangle with vertices at 0, z and w is equilateral.

- (b) Prove that if $x > 0$, $y > 0$ and $x + y = 1$, then **2**

$$\left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) \geq 9$$

- (c) Let $I_n = \int \frac{\sin nx}{\cos x} dx$ for integers $n \geq 0$. **3**

Show that

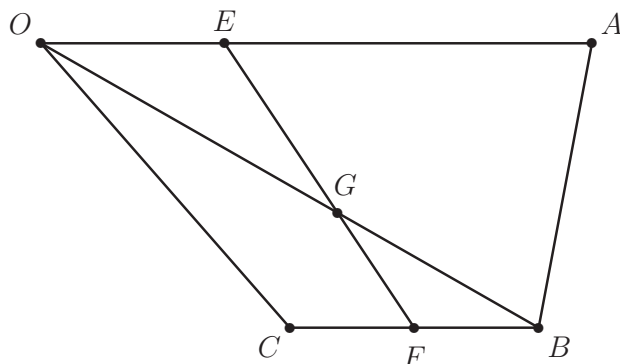
$$I_n = \frac{2}{1-n} \cos((n-1)x) - I_{n-2}, \text{ for } n \geq 2.$$

- (d) Prove by mathematical induction that for all integers $n \geq 0$, **3**

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^n} \leq 1 + n.$$

Examination continues overleaf...

- (e) The sides OA and CB of a trapezium $OABC$ are parallel, with $OA > CB$. The point E on OA is such that $OE : EA = 1 : 2$, and F is the midpoint of CB . The point G is the intersection of OB and EF .



Let $\underline{a}$ and $\underline{c}$ be the position vectors of points A and C respectively.

- (i) Show that B has position vector $\lambda \underline{a} + \underline{c}$ for some scalar λ . **1**
- (ii) Hence, or otherwise, show that the position vector of G is **3**

$$\left(\frac{2\lambda}{2+3\lambda} \right) \underline{a} + \left(\frac{2}{2+3\lambda} \right) \underline{c}.$$

End of Question 15

Question 16 (15 marks) Start on a NEW page**Marks**

- (a) Suppose that a , b and c are the lengths of a right-angled triangle, where c is the hypotenuse. **3**

Show that if $n \leq 2$, then $c^n \leq a^n + b^n$.

- (b) A particle is projected vertically upwards from the Earth's surface with initial speed $U \text{ ms}^{-1}$. **4**

Experiments show that the acceleration due to gravity of an object at a distance x from the centre of the Earth is proportional to the inverse square of the distance x , towards the centre of the Earth; i.e.

$$a = -\frac{k}{x^2},$$

for some positive constant k . (Do NOT prove this.)

Show that the distance travelled by the particle before it has lost three-quarters of its initial speed is

$$\frac{15RU^2}{32gR - 15U^2} \text{ m},$$

where R is the radius of the Earth and g is the acceleration at the Earth's surface.

- (c) Let $I_n = \int_0^\alpha (\sec x + \tan x)^n dx$, where n is a non-negative integer and $0 < \alpha < \frac{\pi}{2}$.

- (i) For $n \geq 1$, show that **3**

$$\frac{1}{2}(I_{n+1} + I_{n-1}) = \frac{1}{n} \left((\sec \alpha + \tan \alpha)^n - 1 \right).$$

- (ii) Deduce that **2**

$$I_n < \frac{1}{n} \left((\sec \alpha + \tan \alpha)^n - 1 \right), \text{ for } n \geq 1.$$

Examination continues overleaf...

- (d) Suppose that z_1 , z_2 and $z_3 \in \mathbb{C}$ satisfy the following conditions simultaneously. **3**

$$\begin{cases} |z_1| = |z_2| = |z_3| = 1 \\ \frac{z_1}{z_2} + \frac{z_2}{z_3} + \frac{z_3}{z_1} = 1 \\ z_2 \neq z_3 \text{ and } z_3 \neq z_1. \end{cases}$$

Prove that if a , b and $c \in \mathbb{R}$, then $|az_1 + bz_2 + cz_3| = \sqrt{a^2 + b^2 + c^2 + 2ab}$.

End of paper

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10.

Questions

Marks

- | | | |
|---|--|---|
| 1 | What is the converse of the statement “if you can’t see my mirrors, then I can’t see you”? | 1 |
| | A. If I can’t see your mirrors, then you can’t see me. | |
| | ☒ B. If I can’t see you, then you can’t see my mirrors. | |
| | C. If you can see me, then I can see your mirrors. | |
| | D. If I can see you, then you can see my mirrors. | |
| | | |
| 2 | What is the contrapositive of the statement “if you can’t see my mirrors, then I can’t see you”? | 1 |
| | A. If I can’t see your mirrors, then you can’t see me. | |
| | B. If I can’t see you, then you can’t see my mirrors. | |
| | C. If you can see me, then I can see your mirrors. | |
| | ☒ D. If I can see you, then you can see my mirrors. | |

- 3 If $\underline{u} = 5\underline{i} + (t-3)\underline{j} + 3\underline{k}$ and $\underline{v} = \underline{i} + (t+9)\underline{j} + 2\underline{k}$ are perpendicular, what are the possible values of t ? 1

A. $t = 8, 2$

B. $t = -8, -2$

C. $t = 8, -2$

D. $t = -8, 2$

$$5 + (t-3)(t+9) + 6 = 0$$

$$t^2 + 6t - 16 = 0$$

$$(t+8)(t-2) = 0$$



- 4 Suppose $\underline{a}$, $\underline{b}$ and $\underline{c}$ are three vectors such that $\underline{a} + \underline{b} + \underline{c} = \underline{0}$, $|\underline{a}| = 3$, $|\underline{b}| = 5$, and $|\underline{c}| = 7$. 1

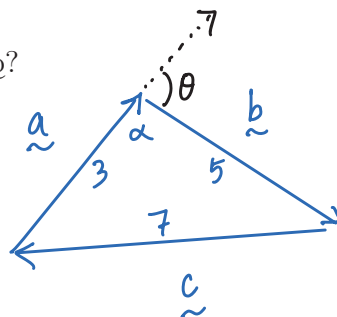
What is the angle between $\underline{a}$ and $\underline{b}$?

A. $\frac{\pi}{6}$

B. $\frac{2\pi}{3}$

C. $\frac{5\pi}{3}$

D. $\frac{\pi}{3}$



$$\cos \alpha = \frac{3^2 + 5^2 - 7^2}{2(3)(5)}$$

$$\alpha = \frac{2\pi}{3} \quad \therefore \theta = \frac{\pi}{3}$$

- 5 A particle of mass m moves so that its velocity v is given by $v = f(x)$. 1

The resultant force that causes this motion is given by:

A. $mf'(x)$

B. $mx f'(x)$

C. $mf' \left(\frac{1}{2}x^2 \right)$

D. $mf(x)f'(x)$

$$F = ma$$

$$= m v \frac{dv}{dx}$$

$$= m f(x) \cdot f'(x)$$



- 6 Let ω be a non-real cube root of unity.

$$\begin{cases} 1 + \omega + \omega^2 = 0 \\ \omega^3 = 1 \end{cases}$$

1

What is the value of $(1 + 2\omega + 3\omega^2)(1 + 3\omega + 2\omega^2)$?

A. 3

B. 6

C. 9

D. 12

$$\begin{aligned} & (1 + \omega + \omega^2 + \omega + 2\omega^2)(1 + \omega + \omega^2 + 2\omega + \omega^2) \\ &= (\omega + 2\omega^2)(2\omega + \omega^2) \\ &= 2\omega^2 + \omega^3 + 4\omega^3 + 2\omega^4 \\ &= 2\omega^2 + 5 + 2\omega \\ &= 2(\omega^2 + \omega) + 5 = 2(-1) + 5 = 3 \end{aligned}$$

- 7 Suppose that z_1 and z_2 are two distinct non-zero complex numbers satisfying $|z_1 + z_2| = |z_1 - z_2|$.

1

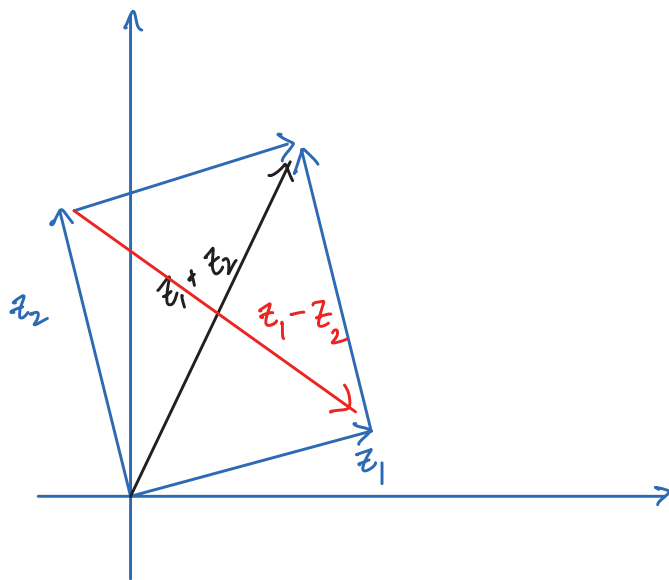
Which of the following is always true?

A. $z_1 z_2$ is purely imaginary.

B. $\frac{z_1}{z_2}$ is purely imaginary.

C. $\frac{z_1 + z_2}{z_1 - z_2}$ is purely imaginary.

D. None of the above.



$$|z_1 + z_2| = |z_1 - z_2|$$

$\therefore$ equal diagonals

$\therefore$ parallelogram must be a rectangle

$$\therefore \text{Arg}(z_2) - \text{Arg}(z_1) = \frac{\pi}{2}$$

$$\therefore \text{Arg}\left(\frac{z_2}{z_1}\right) = \frac{\pi}{2}, \text{ i.e. } \frac{z_2}{z_1} \text{ is purely imaginary}$$

$$\therefore \frac{z_1}{z_2} \text{ is purely imaginary.}$$

- 8 A particle moves from the initial conditions $t = t_0$, $x = x_0$, $v = v_0$ to the final conditions $t = t_F$, $x = x_F$, $v = v_F$ under a force that produces an acceleration of $\ddot{x}$. 1

Which of the following is incorrect?

A. If $\ddot{x} = f(t)$, then $v_F = \int_{t_0}^{t_F} f(t) dt + v_0$.

B. If $\ddot{x} = f(v)$, then $x_F = \int_{v_0}^{v_F} \frac{f(v)}{v} dv + x_0$.

C. If $\ddot{x} = f(v)$, then $t_F = \int_{v_0}^{v_F} \frac{1}{f(v)} dv + t_0$.

D. If $\ddot{x} = f(x)$, then $(v_F)^2 = 2 \int_{x_0}^{x_F} f(x) dx + (v_0)^2$.

$$\frac{v dv}{dx} = f(v)$$

$$\int_{v_0}^{v_F} \frac{v dv}{f(v)} = \int_{x_0}^{x_F} dx$$

- 9 Suppose that a , b and c are non-zero real numbers satisfying 1

$$\frac{a^2}{b^2 + c^2} < \frac{b^2}{c^2 + a^2} < \frac{c^2}{a^2 + b^2}.$$

Which of the following statements can be deduced from the preceding inequality?

A. $a < b < c$

B. $|a| < |b| < |c|$

C. $c < b < a$

D. $|c| < |b| < |a|$

$$\frac{b^2 + c^2}{a^2} > \frac{c^2 + a^2}{b^2} > \frac{a^2 + b^2}{c^2}$$

Add 1 to each side

$$\frac{a^2 + b^2 + c^2}{a^2} > \frac{a^2 + b^2 + c^2}{b^2} > \frac{a^2 + b^2 + c^2}{c^2}$$

$$\therefore \frac{1}{a^2} > \frac{1}{b^2} > \frac{1}{c^2}$$

$$\therefore a^2 < b^2 < c^2 \quad \therefore |a| < |b| < |c|$$

- 10 Three non-zero vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ on a number plane are such that $\underline{b} \cdot \underline{c} = 2|\underline{a}|$, $\underline{a} \cdot \underline{c} = 3|\underline{b}|$, and $\underline{a}$ is perpendicular to $\underline{b}$. 1

What is the minimum value of $|\underline{c}|$?

A. $2\sqrt{2}$

B. $2\sqrt{3}$

C. $2\sqrt{5}$

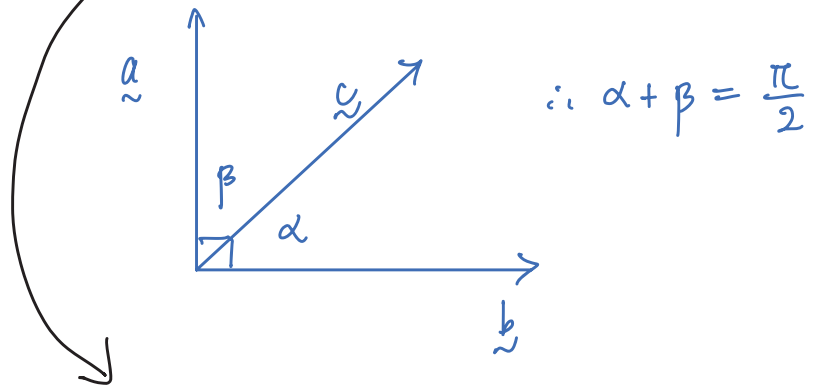
D. None of the above.

$$|\underline{b}| |\underline{c}| \cos \alpha = 2 |\underline{a}| \quad (\text{note: } 0 \leq \alpha \leq \pi)$$

$$\therefore \cos \alpha > 0 \quad \therefore \alpha \text{ is acute}$$

Similarly,

$$|\underline{a}| |\underline{c}| \cos \beta = 3 |\underline{b}| \quad \therefore \beta \text{ is acute}$$



$$\therefore |\underline{a}| |\underline{c}| \sin \alpha = 3 |\underline{b}|$$

$$\text{Now } |\underline{c}| \sin \alpha = \frac{3 |\underline{b}|}{|\underline{a}|}$$

Examination continues overleaf...

$$\text{and } |\underline{c}| \cos \alpha = \frac{2 |\underline{a}|}{|\underline{b}|}$$

$$\therefore |\underline{c}|^2 = \frac{9 |\underline{b}|^2}{|\underline{a}|^2} + \frac{4 |\underline{a}|^2}{|\underline{b}|^2}$$

$$= \frac{9 |\underline{b}|^4 + 4 |\underline{a}|^4}{|\underline{a}|^2 |\underline{b}|^2}$$

$$\geq \frac{2(3 |\underline{b}|^2)(2 |\underline{a}|^2)}{|\underline{a}|^2 |\underline{b}|^2}$$

$$= 12$$

$$\therefore |\underline{c}| \geq 2\sqrt{3}$$

Question 11 (15 marks) Start on a NEW page

- (a) Give a counterexample to disprove the following statement:

For all $x, y \in \mathbb{R}$, if $x^2 \geq y^2$, then $x \geq y$.

Suppose $x = -1, y = 0$
 $(-1)^2 \geq 0 \checkmark$ but $-1 \not\geq 0 \checkmark$
 $\therefore$ This disproves the statement.

Comments:

Students have to clearly disprove the statement through a conclusion.

- (b) Write down the negation of the following statement:

There exists $k \in \mathbb{R}$ such that $k^2 < 0$.

$\sim (\exists k \in \mathbb{R}, k^2 < 0) \equiv \forall k \in \mathbb{R}, k^2 \geq 0 \checkmark$

Comments:

Some students left their statement as an existential statement and failed to include zero in their inequality.

- (c) Consider the vectors $\underline{u} = 3\hat{i} + 2\hat{j} + \hat{k}$, $\underline{v} = -5\hat{j} + 2\hat{k}$ and $\underline{w} = 2\hat{i} + 2\hat{j} + 5\hat{k}$.

- (i) Prove that $\underline{v}$ and $\underline{w}$ are perpendicular.

RTP: $\underline{v} \cdot \underline{w} = 0$

$$\text{LHS} = \begin{pmatrix} 0 \\ -5 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 2 \\ 5 \end{pmatrix} = 2 \times 0 - 5 \times 2 + 2 \times 5 = 0$$

 Since $\underline{v} \cdot \underline{w} = 0$, $\underline{v} \perp \underline{w} \checkmark$

Comments:

Done well.

- (ii) Verify that $|\underline{u} + \underline{v} + \underline{w}| \leq |\underline{u}| + |\underline{v}| + |\underline{w}|$.

$$\text{LHS} = \left| \begin{pmatrix} 3+0+2 \\ 2-5+2 \\ 1+2+5 \end{pmatrix} \right| = \left| \begin{pmatrix} 5 \\ -1 \\ 8 \end{pmatrix} \right| = \sqrt{25+1+64} = \sqrt{90} \doteq 9.49 \checkmark$$

$$\text{RHS} = \sqrt{9+4+1} + \sqrt{25+4} + \sqrt{4+4+25} = \sqrt{14} + \sqrt{29} + \sqrt{33} \doteq 14.87$$

$\therefore \text{LHS} \leq \text{RHS} \checkmark$
 OR $\therefore \underbrace{|\underline{u} + \underline{v} + \underline{w}|}_{\doteq 9.49} \leq \underbrace{|\underline{u}| + |\underline{v}| + |\underline{w}|}_{\doteq 14.87}$

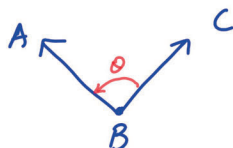
Mark	Criteria
2	Correct solution.
1	Finds $ \underline{u} + \underline{v} + \underline{w} $ and $ \underline{u} + \underline{v} + \underline{w} $, but fails to clearly show that one is greater than the other through correct decimal rounding.

Comments:

Students had to **clearly** show this inequality was true by decimal rounding. Students that proved the result using the triangle identity were awarded with full marks.

- (d) Given the points $A(1, 3, 5)$, $B(2, -4, 11)$ and $C(-3, -1, 0)$, find $\angle ABC$.
Give your answer correct to the nearest minute.

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|}$$



$$\begin{aligned}\vec{BA} &= \vec{OA} - \vec{OB} = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 11 \end{pmatrix} \\ &= \begin{pmatrix} -1 \\ 7 \\ -6 \end{pmatrix}\end{aligned}$$

$$\begin{aligned}\text{Similarly, } \vec{BC} &= \vec{OC} - \vec{OB} = \begin{pmatrix} -3 \\ -1 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ -4 \\ 11 \end{pmatrix} \\ &= \begin{pmatrix} -5 \\ 3 \\ -11 \end{pmatrix}\end{aligned}$$

$$\therefore \cos \theta = \frac{\vec{BA} \cdot \vec{BC}}{|\vec{BA}| |\vec{BC}|} = \frac{5 + 21 + 66}{\sqrt{1+49+36} \sqrt{25+9+121}} = \frac{92}{\sqrt{86} \sqrt{155}}$$

$$\therefore \theta = 37^{\circ} 10'$$

Mark	Criteria
3	Correct solution.
2	Makes error in their final calculation or in obtaining their vectors $\vec{BA}$, $\vec{BC}$.
1	Makes 2 subsequent errors in finding their vectors and in calculating their angle.

Comments:

Students were not deducted for rounding. Many students made mistakes with finding $\vec{CB}$ instead of $\vec{BC}$. Students made errors in calculating their angle by accidentally omitting the square roots or mistyping their expression in their calculator.

- (e) By using integration by parts, evaluate $\int_0^{\pi/2} x \cos x \, dx$.

$$\text{let } u = x \quad u' = 1$$

$$v' = \cos x \quad v = \sin x$$

$$I = \left[x \sin x \right]_0^{\pi/2} - \int_0^{\pi/2} 1 \cdot \sin x \, dx$$

$$= \frac{\pi}{2} (1) - 0 + \left[\cos x \right]_0^{\pi/2}$$

$$= \frac{\pi}{2} + (0 - 1) = \frac{\pi}{2} - 1$$

Mark	Criteria
3	Correct solution.
2	Makes error in their integration by parts.
1	Makes 2 subsequent errors in integration by parts.

Comments:

Most students obtained at least 2 marks. Some errors that occurred included mistakes with the negative and forgetting to substitute the zero into the antiderivative.

- (f) Evaluate $\text{Arg}(i^{2025})$, where $\text{Arg}(z)$ denotes the principal argument of z .

$$\arg(i^{2025}) = 2025 \times \frac{\pi}{2}$$

$$= \frac{2025\pi}{2}$$

$$= \pi(1012 + 0.5)$$

$$= \underbrace{1012\pi}_{2k\pi} + \frac{\pi}{2}$$

$$\Rightarrow \text{Arg}(i^{2025}) = \frac{\pi}{2}$$

Note: $\arg z^n = n \arg z$

Mark	Criteria
2	Correct solution.
1	Makes substantial progress or makes an error.

Comments:

Done well.

- (g) Prove that $\sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$.

$$\text{let } z = e^{i\theta} = \cos \theta + i \sin \theta \quad (1)$$

$$\begin{aligned} z^{-1} &= e^{-i\theta} = \cos(-\theta) + i \sin(-\theta) \\ &= \cos \theta - i \sin \theta \quad (2) \end{aligned}$$

$$(1) - (2) \Rightarrow e^{i\theta} - e^{-i\theta} = 2i \sin \theta \quad [\div 2i]$$

$$\therefore \sin \theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$$

Comments:

Done well.

Q12

$$a. \int \cos^3 x \sin^2 x \, dx$$

$$= \int \sin^2 x \cos^2 x \cos x \, dx$$

$$= \int \sin^2 x (1 - \sin^2 x) \cos x \, dx$$

$$= \int (\sin^2 x \cos x - \sin^4 x \cos x) \, dx$$

$$= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C$$

Alternate Method: let $u = \sin x$

$$\frac{du}{dx} = \cos x$$

$$du = \cos x \, dx$$

$$I = \int \sin^2 x (1 - \sin^2 x) \cos x \, dx$$

$$= \int (u^2 - u^4) \, du \quad \checkmark$$

$$= \frac{1}{3} u^3 - \frac{1}{5} u^5 + C$$

$$= \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C \quad \checkmark$$

correct working and
answer : AW 2

correct integrand in
terms of u : AW 1

Q₁₂

b. $\int \frac{d\theta}{3 + 2\cos 4\theta}$

Let $t = \tan 2\theta$

$$\frac{dt}{d\theta} = 2\sec^2 \theta d\theta$$

$$d\theta = \frac{dt}{2(t^2 + 1)} \quad \checkmark$$

$$I = \int \frac{\frac{dt}{2(t^2 + 1)}}{3 + \frac{2(1-t^2)}{1+t^2}}$$

$$I = \frac{1}{2} \int \frac{dt}{t^2 + 5} \quad \checkmark$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{5}} \tan^{-1}\left(\frac{t}{\sqrt{5}}\right)$$

$$= \frac{1}{2\sqrt{5}} \tan^{-1}\left(\frac{\tan 2\theta}{\sqrt{5}}\right) + C \quad \checkmark$$

Express $d\theta$ in terms of t

AW: 1

Correct integrand in terms of t AW: 2

Correct working and Ans AW: 3

C/i

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3\cos^2 \theta \sin \theta i + 3\cos i \sin^2 \theta + i^3 \sin^3 \theta$$

$$= \cos^3 \theta - 3\cos \theta (1 - \cos^2 \theta) + (\sin(1 - \sin^2 \theta) - \sin^3 \theta) i$$

$$= 4\cos^3 \theta - 3\cos \theta + (\sin \theta - 2\sin^3 \theta) i \quad \checkmark \quad (1)$$

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta \quad (2) \quad \text{De Moivre's}$$

Equating real part from (1) and (2)

$$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta \quad \checkmark$$

use binomial expansion correctly AW: 1

C/ii let $x = \cos \theta$: $8x^3 - 6x + 1 = 0$

$$2(4x^3 - 3x) = -1$$

$$4x^3 - 3x = -\frac{1}{2}$$

$$4\cos^3 \theta - 3\cos \theta = -\frac{1}{2} \quad \checkmark$$

$$\cos 3\theta = -\frac{1}{2} \quad \text{from (i)}$$

$$3\theta = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}, 3\pi - \frac{\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9} \quad \checkmark$$

The solutions: $x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9} \quad \checkmark$

Express $\cos 3\theta = -\frac{1}{2}$ AW: 1

Having $\cos 3\theta = -\frac{1}{2}$ and $\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$ AW: 2

provide correct working and solutions AW: 3

C/iii $8x^3 - 6x + 1 = 0$

Sum of roots: $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = -\frac{b}{a} = 0$

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \left(\pi - \frac{\pi}{9}\right) = 0 \quad \checkmark$$

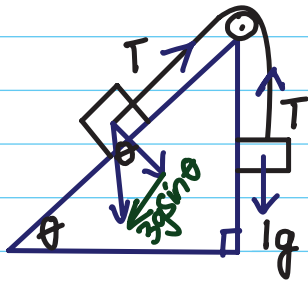
$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} - \cos \frac{\pi}{9} = 0$$

$$\cos \frac{\pi}{9} - \cos \frac{2\pi}{9} - \cos \frac{4\pi}{9} = 0 \quad \checkmark$$

• use sum of the roots $= -\frac{b}{a} = 0$ AW: 1

• Express $\cos \frac{8\pi}{9} = -\cos \frac{\pi}{9}$: AW 1

12/d



The system is in equilibrium $\underline{a} = 0$

Forces act on 3 kg object

$$T = 3g \sin \theta \quad \checkmark$$

Forces act on 1 kg object

$$T = 1g \quad \checkmark$$

$$T = T : \quad 3g \sin \theta = 1g$$

$$\sin \theta = \frac{1}{3}$$

$$\theta = 19.5^\circ \quad \checkmark$$

provide $T = 3g \sin \theta$ OR $T = 1g$ AW 1

solve equation and provide $\sin \theta = \frac{1}{3}$ AW 2

provide correct working and solution AW 3

Question 13

(a)

(i) $\frac{1}{2}v^2 = 6 + 4x - 2x^2$ ✓

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$$

$$= 4 - 4x$$

$$= -4(x-1) \quad \checkmark$$

∴ particle is undergoing SHM.

Comment

- Generally well done
- Students that have shown that

$$v^2 = n^2(a^2 - (x-c)^2)$$

were not awarded any marks (as this makes (a)(ii) very easy and the marks are awarded there).

⎧ To prove a particle is undergoing SHM,
⎩ $\ddot{x} = -n^2(x-c)$ must be shown.

$$(ii) \quad n = 2, \quad \therefore \text{period} = \frac{2\pi}{2} = \pi \quad \checkmark$$

C.O.M, $x = 1$

$$\text{Let } v = 0$$

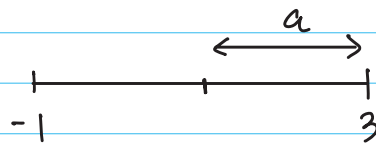
$$-4x^2 + 8x + 12 = 0$$

$$x^2 - 2x - 3 = 0$$

$$(x - 3)(x + 1) = 0$$

$$x = 3, -1$$

$$\therefore a = \frac{3 - (-1)}{2}$$
$$= 2 \quad \checkmark$$



Comment

- Generally well done

(b) Assume that $\sqrt[3]{5} = \frac{p}{q}$, where p and q are coprime positive integers.

$$5 = \frac{p^3}{q^3}$$

$$5q^3 = p^3$$

$\therefore 5$ is a factor of p^3 and hence
a factor of p
(since 5 is prime)

✓
better to include,
however it was
not penalised
this time if
students did not
write it.

$$\therefore p = 5k, k \in \mathbb{Z}$$

$$\therefore 5q^3 = 125k^3$$

$$q^3 = 25k^3 = 5(5k^3)$$

$\therefore 5$ is a factor of q^3 , and hence
a factor of q

This contradicts the assumption that
 p and q are coprime

$\therefore \sqrt[3]{5}$ must be irrational.

$$(c) \quad \operatorname{Re}(z^2) \leq (\operatorname{Re} z)^2$$

$$\text{Let } z = x + iy$$

$$\operatorname{Re}(x^2 + 2ixy - y^2) \leq x^2$$

$$x^2 - y^2 \leq x^2$$

$$-y^2 \leq 0$$

$$\therefore y^2 \geq 0$$

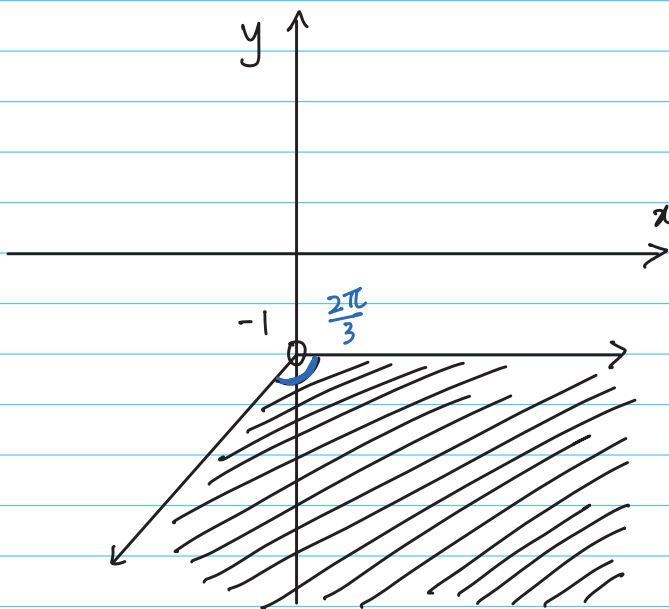
$\therefore$ all real y ✓

Must write

Must show

AND

$$-\frac{2\pi}{3} \leq \operatorname{Arg}(z - (-i)) \leq 0$$



$$(d) \int_1^{64} \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx$$

$$u^6 = x$$

$$6u^5 du = dx$$

$$x=1, u=1$$

$$x=64, u=2$$

$$= \int_1^2 \frac{6u^5 du}{u^3 + u^2}$$

$$= 6 \int_1^2 \frac{u^3}{u+1} du$$

$$= 6 \int_1^2 \frac{u^3 + 1}{u+1} - \frac{1}{u+1} du$$

Some students
had difficulty
in obtaining
this integrand

$$= 6 \int_1^2 \frac{u^3 - u + 1}{u+1} - \frac{1}{u+1} du$$

$$\left[u^3 + 1 = (u+1)(u^2 - u + 1) \right]$$

$$= 6 \left[\frac{u^3}{3} - \frac{u^2}{2} + u - \ln(u+1) \right]_1^2$$

$$= 6 \left[\left(\frac{8}{3} - 2 + 2 - \ln 3 \right) - \left(\frac{1}{3} - \frac{1}{2} + 1 - \ln 2 \right) \right]$$

$$= 11 + 6 \ln \frac{2}{3}$$

(e) Note that $\frac{x^2 + 1}{x^4 + 8x^2 + 12}$

$$= \frac{y + 1}{y^2 + 8y + 12} \quad \text{where } y = x^2 \quad (\star)$$

$$= \frac{y + 1}{(y + 6)(y + 2)}$$

$$= \frac{\frac{5}{4}}{y + 6} + \frac{-\frac{1}{4}}{y + 2}$$

✓ can be obtained easily with the cover up rule

$$\therefore I = \int \frac{\frac{5}{4}}{x^2 + 6} - \frac{\frac{1}{4}}{x^2 + 2} dx$$

$$= \frac{5}{4\sqrt{6}} \tan^{-1}\left(\frac{x}{\sqrt{6}}\right) - \frac{1}{4\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C$$

Comment

- Only a handful of students recognised that the integrand can be simplified immensely as shown in $(\star)$ above.
- Majority of the students' partial fractions were

$$\frac{Ax + B}{x^2 + 6} + \frac{Cx + D}{x^2 + 2}.$$

This was time costly and susceptible to errors.

Q14.

a/i. $S_1: x^2 + y^2 + z^2 = 9$
 $S_2: (x-3)^2 + (y+\sqrt{6})^2 + (z-7)^2 = 25$

Let d be the distance between the centres of the spheres.

$$d = \sqrt{(3-0)^2 + (-\sqrt{6}-0)^2 + (7-0)^2} \quad \checkmark$$
$$d = 8$$

The sum of two radii $= 3 + 5 = 8 \quad \checkmark$

Since the sum of two radii is the same as the distance between the centres of the two spheres thus the spheres must touch each other at a single point or S_1 and S_2 are externally tangent. $\checkmark$

- use the formula to provide the correct answer for the distance between two spheres Aw: 1
- Having both the sum of the two radii and the distance between the centres of two spheres Aw: 2
- correct working and answer Aw: 3

a/ii $C_1 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad C_2 = \begin{pmatrix} 3 \\ -\sqrt{6} \\ 7 \end{pmatrix}$

$$\overrightarrow{C_1C_2} = \begin{pmatrix} 3 \\ -\sqrt{6} \\ 7 \end{pmatrix} \quad \checkmark$$

$$l_1: \underline{r} = \lambda \underline{d} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -\sqrt{6} \\ 7 \end{pmatrix}$$

$$l_1: \underline{r} = \lambda \begin{pmatrix} 3 \\ -\sqrt{6} \\ 7 \end{pmatrix} \quad \checkmark$$

provide vector direction of l_1 : Aw 1

Q14
a/iii $l_2: \underline{r}_2 = \begin{pmatrix} 1 \\ -\sqrt{6} \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$

$$l_2: \underline{r}_1 = \lambda \begin{pmatrix} -\frac{3}{\sqrt{6}} \\ 7 \end{pmatrix}$$

$$3\lambda = 1 - \lambda \quad (1)$$

$$-\sqrt{6}\lambda = -\sqrt{6} + \lambda \quad (2) \quad \checkmark$$

$$7\lambda = 2 + 2\lambda \quad (3)$$

(1) + (2)

$$\lambda(3 - \sqrt{6}) = 1 - \sqrt{6}$$

$$\lambda = \frac{1 - \sqrt{6}}{3 - \sqrt{6}} = \frac{-3 - 2\sqrt{6}}{3} \text{ subs into (1)}$$

$$3\left(\frac{-3 - 2\sqrt{6}}{3}\right) = 1 - \lambda$$

$$\lambda = 4 + 2\sqrt{6}$$

Note: the values of λ and μ can vary ($\lambda = \frac{4}{13}, \mu = \frac{1}{13}$)

Subs λ and μ into (3)

$$7\left(\frac{-3 - 2\sqrt{6}}{3}\right) = 2 + 2(4 + 2\sqrt{6})$$

$$\frac{-21 - 14\sqrt{6}}{3} \neq 10 + 4\sqrt{6} \quad \checkmark$$

l_1 and l_2 do not intersect. No points of intersection.

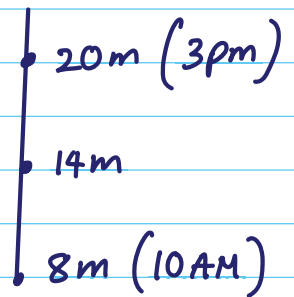
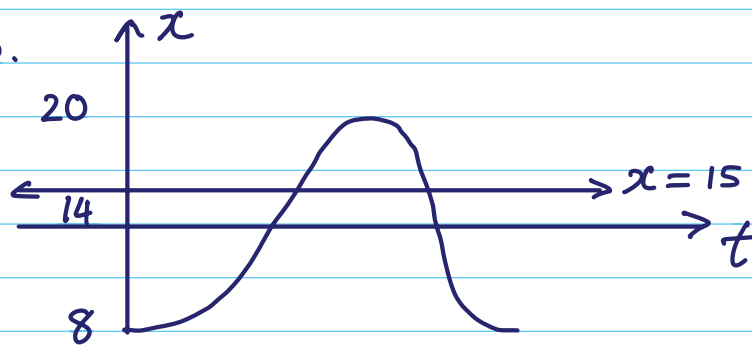
a/iv.

Since l_1 and l_2 do not intersect from (iii) and direction vector of l_1 and l_2 are Not multiple scalar of each other

$$\begin{pmatrix} 3 \\ -\sqrt{6} \\ 7 \end{pmatrix} \neq a \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}, a \in \mathbb{R} \therefore l_1 \nparallel l_2 \quad \checkmark$$

1M for stating l_1 and l_2 not intersecting and another mark for stating l_1 is not $\parallel l_2$.

Q14/b.



Centre of motion $c = \frac{20+8}{2} = 14$

amplitude: $a = 20 - 14 = 14 - 8 = 6$ ✓

Period: $T = \frac{2\pi}{n} = \frac{2\pi}{n} = 10 \rightarrow n = \frac{\pi}{5}$

$x = a \cos(n t + \alpha) + c$
At $t = 0$ (8AM), $x = 8$ m

$8 = 6 \cos \alpha + 14$

$\cos \alpha = -1 \rightarrow \alpha = \pi$

$x = 6 \cos\left(\pi + \frac{\pi}{5}t\right) + 14$

$x = -6 \cos\left(\frac{\pi}{5}t\right) + 14$ ✓

Ship requires minimum of 15m depth.

$15 = -6 \cos\left(\frac{\pi}{5}t\right) + 14$

$\cos \frac{\pi}{5}t = -\frac{1}{6}$

$t = \frac{\pi - \cos^{-1}\left(\frac{1}{6}\right)}{\pi/5} = 2.7665 \text{ hrs}$

$= 2 \text{ hrs and } 46 \text{ mins}$ ✓

$\therefore$ Earliest time 10AM + 2hrs, 46mins = 12:46PM

provide either amplitude, centre of motion or $n = \frac{\pi}{5}$ AW: 1
Correct equation: $x = -6 \cos\left(\frac{\pi}{5}t\right) + 14$ AW: 2
correct working and solution AW: 3

Q14/c.

Let the two consecutive numbers be:

$$a, b = a + 1$$

$$c = ab = a(a + 1)$$

$$(a^2 + a + 1)^2 = a^4 + 2a^2(a + 1) + (a + 1)^2$$

$$= a^4 + 2a^3 + 2a^2 + a^2 + 2a + 1$$

$$= a^4 + 2a^3 + 3a^2 + 2a + 1 \quad (*) \checkmark$$

$$\sqrt{a^2 + b^2 + c^2} = \sqrt{a^2 + (a + 1)^2 + a^2(a + 1)^2}$$

$$= \sqrt{a^2 + a^2 + 2a + 1 + a^2(a^2 + 2a + 1)}$$

$$= \sqrt{a^4 + 2a^3 + 3a^2 + 2a + 1}$$

$$= \sqrt{(a^2 + a + 1)^2} \quad \text{from } (*)$$

$$= a^2 + a + 1 \quad \checkmark$$

$$= a(a + 1) + 1 \quad \text{This is odd.}$$

Even + 1 = odd

$a(a + 1)$ = the product of two consecutive numbers is always even plus 1, gives

$$(a^2 + a + 1) \text{ odd.}$$

$$\therefore \sqrt{a^2 + b^2 + c^2} = a^2 + a + 1 \text{ is an odd integer.} \quad \checkmark$$

Expand $(a^2 + a + 1)^2$ correctly: AW 1

Express $\sqrt{a^2 + b^2 + c^2} = a^2 + a + 1$: AW 2M

Correct proof: AW 3

Question Q15(a)	Marks
· Provides correct solution.	3 marks
· Proves $ z - w ^2 = z ^2 - (z\bar{w} + w\bar{z}) + w ^2$.	2 marks
· Proves that $ z = w $ and attempts to show $ z - w = z $.	1 mark

Solution (Method 1 of 4, Sum of two squares identity)

The triangle with vertices at 0, z and w has sides of length $|z|$, $|w|$ and $|z - w|$.

Since $|z|^2 = |w|^2$ we have $|z| = |w|$.

Hence, it suffices to prove that $|z - w| = |z|$, because then $|z - w| = |z| = |w|$ and the triangle is equilateral as required. ✓

Consider $|z - w|^2$, and use the fact that for any complex number u , $|u|^2 = u\bar{u}$:

$$\begin{aligned}
 |z - w|^2 &= (z - w)\overline{(z - w)} \\
 &= (z - w)(\bar{z} - \bar{w}) \\
 &= z\bar{z} - z\bar{w} - w\bar{z} + w\bar{w} \\
 &= |z|^2 - (z\bar{w} + w\bar{z}) + |w|^2 \quad \checkmark \\
 &= |z|^2 + |w|^2 - (z\bar{w} + \overline{z\bar{w}}) \\
 &= |z|^2 + |w|^2 - 2\operatorname{Re}(z\bar{w})
 \end{aligned}$$

Now use the given information that $|w|^2 = |z|^2$ and $2\operatorname{Re}(z\bar{w}) = |z|^2$:

$$\begin{aligned}
 |z - w|^2 &= |z|^2 + |z|^2 - |z|^2 \\
 &= |z|^2
 \end{aligned}$$

which implies $|z - w| = |z|$, which is what we were aiming to show. ✓

Question 15(a) (continued)

Solution (Method 2 of 4, Cartesian form)

Let $z = a + ib$ and let $w = x + iy$. Then

$$\begin{aligned}\operatorname{Re}(z\bar{w}) &= \operatorname{Re}((a + ib)(x - iy)) \\ &= \operatorname{Re}(ax + by + ibx - iay) \\ &= ax + by \quad \checkmark\end{aligned}$$

So the given information $|z|^2 = |w|^2 = 2\operatorname{Re}(z\bar{w})$ becomes $a^2 + b^2 = x^2 + y^2 = 2ax + 2by$ (*).

$$\begin{aligned}|z - w|^2 &= |(a - x) + i(b - y)|^2 \\ &= (a - x)^2 + (b - y)^2 \\ &= a^2 - 2ax + x^2 + b^2 - 2by + y^2 \\ &= a^2 + b^2 + x^2 + y^2 - 2(ax + by) \quad \checkmark \\ &= a^2 + b^2 + a^2 + b^2 - (a^2 + b^2) \quad \text{using (*)} \\ &= a^2 + b^2 \\ &= |z|^2\end{aligned}$$

Hence $|z - w|^2 = |z|^2 = |w|^2$ and so $|z - w| = |z| = |w|$, which implies all sides are equal and the triangle is equilateral. $\checkmark$

Question 15(a) (continued)

Solution (Method 3 of 4, Mod-arg form)

Since $|z|^2 = |w|^2$, we have $|z| = |w|$.

Hence, let $z = r \operatorname{cis} \theta$ and $w = r \operatorname{cis} \phi$.

Since $2\operatorname{Re}(z\bar{w}) = |z|^2$, we have

$$2\operatorname{Re}(z\bar{w}) = r^2 \checkmark$$

$$2\operatorname{Re}(r^2 \operatorname{cis} \theta \operatorname{cis} (-\phi)) = r^2$$

$$2\operatorname{Re}(r^2 \operatorname{cis} (\theta - \phi)) = r^2$$

$$\operatorname{Re}(\operatorname{cis} (\theta - \phi)) = \frac{1}{2}$$

$$\cos(\theta - \phi) = \frac{1}{2}$$

$$\theta - \phi = \pm \frac{\pi}{3} \checkmark$$

Therefore the angle between z and w is $\frac{\pi}{3}$.

Since $|z| = |w|$, the triangle is isosceles, but by angle sum of a triangle, all angles are $\frac{\pi}{3}$,

which implies the triangle is equilateral. $\checkmark$

Question 15(a) (continued)

Solution (Method 4 of 4, Vectors)

As in Method 2, let $z = a + ib$ and $w = x + iy$ so that $\operatorname{Re}(z\bar{w}) = ax + by$. ✓

If θ is the angle between the vectors $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} x \\ y \end{pmatrix}$, then

$$\begin{aligned}\cos \theta &= \frac{\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix}}{\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| \left| \begin{pmatrix} x \\ y \end{pmatrix} \right|} \\ &= \frac{ax + by}{\sqrt{a^2 + b^2} \sqrt{x^2 + y^2}} \quad \checkmark \\ &= \frac{ax + by}{a^2 + b^2} \text{ since } a^2 + b^2 = x^2 + y^2 \\ &= \frac{ax + by}{2(ax + by)} \text{ since } |z|^2 = 2\operatorname{Re}(z\bar{w}) \\ &= \frac{1}{2}\end{aligned}$$

Hence $\theta = \frac{\pi}{3}$, which implies the angle between z and w is $\frac{\pi}{3}$.

As in Method 3, this implies the triangle is equilateral. ✓

Comments

- This was a rich question with many successful approaches, as shown above.
- There were significant notation mistakes that were not penalised.
- For example, $\overrightarrow{zw}$ instead of $\overrightarrow{ZW}$.
- Another example: $|zw|$ instead of $|ZW|$. This was a very confusing error of notation.
- The correct expression was $|z - w|$ which is very different from $|zw| = |z \times w|$.

Question Q15(b)	Marks
· Provides correct solution.	2 marks
· Obtains $\text{LHS} = 1 + \frac{2}{xy}$ or manages to prove $\text{LHS} \geq 8$	1 mark

Solution

Expand the brackets:

$$\begin{aligned}
 \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) &= 1 + \frac{1}{x} + \frac{1}{y} + \frac{1}{xy} \\
 &= 1 + \frac{x+y}{xy} + \frac{1}{xy} \\
 &= 1 + \frac{1}{xy} + \frac{1}{xy} \\
 &= 1 + \frac{2}{xy}, \quad \checkmark
 \end{aligned}$$

where we used $x + y = 1$. We will now show $\frac{2}{xy} \geq 8$, which will imply the result.

$$\begin{aligned}
 (x - y)^2 &\geq 0 \\
 x^2 + y^2 &\geq 2xy \\
 x^2 + 2xy + y^2 &\geq 4xy \\
 (x + y)^2 &\geq 4xy \\
 1^2 &\geq 4xy \quad \text{since } x + y = 1 \\
 \frac{1}{xy} &\geq 4 \\
 \frac{2}{xy} &\geq 8
 \end{aligned}$$

Therefore we have

$$\begin{aligned}
 \left(1 + \frac{1}{x}\right) \left(1 + \frac{1}{y}\right) &= 1 + \frac{2}{xy} \\
 &\geq 1 + 8 \\
 &= 9 \quad \checkmark
 \end{aligned}$$

Comments:

- Expanding the brackets was a simple but efficient approach.
- Many students launched into AM-GM, but this was not well suited to this problem.
- Students are encouraged to try simple approaches first before complicating matters.

Question Q15(c)	Marks
· Provides correct solution.	3 marks
· Applies products to sums with $\sin B \cos A$ where $A = (n-1)x$ and $B = x$.	2 marks
· Applies angle expansion formula for $\sin$ with $A = (n-1)x$ and $B = x$.	1 mark

Solution (Method 1 of 2, Slow but easier to find)

$$\begin{aligned}
I_n &= \int \frac{\sin nx}{\cos x} dx \\
&= \int \frac{\sin((n-1)x + x)}{\cos x} dx \quad (\text{now use angle expansion formula}) \\
&= \int \frac{\sin((n-1)x) \cos x + \sin x \cos((n-1)x)}{\cos x} dx \quad \checkmark \\
&= \int \frac{\sin((n-1)x) \cos x}{\cos x} dx + \int \frac{\sin x \cos((n-1)x)}{\cos x} dx \\
&= \int \sin((n-1)x) dx + \int \frac{\sin x \cos((n-1)x)}{\cos x} dx \\
&= -\frac{1}{n-1} \cos((n-1)x) + \int \frac{\sin x \cos((n-1)x)}{\cos x} dx \quad (\text{now use products to sums}) \\
&= -\frac{1}{n-1} \cos((n-1)x) + \frac{1}{2} \int \frac{\sin((n-1)x + x) - \sin((n-1)x - x)}{\cos x} dx \quad \checkmark \\
&= -\frac{1}{n-1} \cos((n-1)x) + \frac{1}{2} \int \frac{\sin(nx) - \sin((n-2)x)}{\cos x} dx \\
&= -\frac{1}{n-1} \cos((n-1)x) + \frac{1}{2} \int \frac{\sin(nx)}{\cos x} dx - \frac{1}{2} \int \frac{\sin((n-2)x)}{\cos x} dx \\
I_n &= -\frac{1}{n-1} \cos((n-1)x) + \frac{1}{2} I_n - \frac{1}{2} I_{n-2} \\
2I_n &= -\frac{2}{n-1} \cos((n-1)x) + I_n - I_{n-2} \\
I_n &= \frac{2}{1-n} \cos((n-1)x) - I_{n-2} \quad \checkmark
\end{aligned}$$

Question 15(c) (continued)

Solution (Method 2 of 2, Fast but harder to find)

$$\begin{aligned} I_n + I_{n-2} &= \int \frac{\sin(nx) + \sin((n-2)x)}{\cos x} dx \quad (\text{now use sums to products}) \checkmark \\ &= \int \frac{2 \sin((n-1)x) \cos x}{\cos x} \checkmark \\ &= 2 \int \sin((n-1)x) \\ &= -\frac{2}{n-1} \cos((n-1)x) \\ &= \frac{2}{1-n} \cos((n-1)x) \end{aligned}$$

Therefore $I_n = \frac{2}{1-n} \cos((n-1)x) - I_{n-2}$ as required. $\checkmark$

Comments:

- It is amazing how much easier it is to consider $I_n + I_{n-2}$ than I_n on its own.
- Students should take note of this trick because it might save time in the HSC exam.

Question Q15(d)	Marks
· Provides correct solution.	3 marks
· Substitutes assumption (*) and correctly lists all terms.	2 marks
· Establishes base case $n = 0$.	1 mark

Solution

Base case: For $n = 0$,

$$\begin{aligned}\text{LHS} &= \frac{1}{2^0} \\ &= 1\end{aligned}$$

$$\begin{aligned}\text{RHS} &= 1 + 0 \\ &= 1\end{aligned}$$

Hence $\text{LHS} \leq \text{RHS}$, since both equal 1. ✓

Assume true for $n = k$:

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} \leq 1 + k \quad (*)$$

Prove true for $n = k + 1$

$$\text{R.T.P: } 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}} \leq 1 + (k + 1)$$

$$\text{LHS} = 1 + \underbrace{\frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k}}_{\text{use } (*)} + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}}$$

$$\leq 1 + k + \underbrace{\frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}}}_{2^{k+1} - (2^k + 1) + 1 \text{ terms}} \quad \checkmark \quad (\text{note that } 2^{k+1} - (2^k + 1) + 1 = 2^k)$$

$$\leq 1 + k + \underbrace{\frac{1}{2^k} + \cdots + \frac{1}{2^k}}_{2^k \text{ terms}} \quad (\text{A smaller denominator makes the number larger})$$

$$\leq 1 + k + 2^k \times \frac{1}{2^k}$$

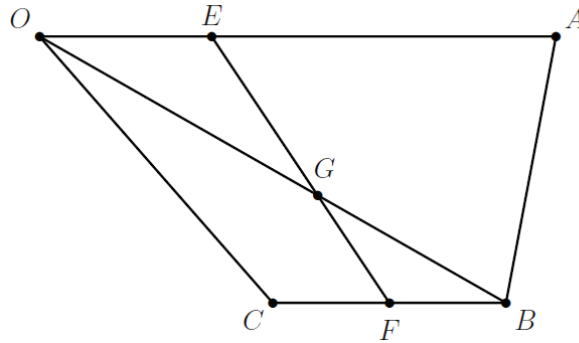
$$= 1 + k + 1$$

$$= \text{RHS} \quad \checkmark$$

Comments:

- Unfortunately, almost all students misunderstood how the series changed for $n = k + 1$.
- They wrote $\text{LHS} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^{k+1}}$.
- However, as above, the correct expression is $\text{LHS} = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}}$
- Consequently, the median mark for this question was 1 out of 3.

Question 15(e)(i)	Marks
· Provides correct solution.	1 mark



Solution

$$\begin{aligned}
 \overrightarrow{OB} &= \overrightarrow{OC} + \overrightarrow{CB} \\
 &= \underline{c} + \lambda \overrightarrow{OA}, \text{ for some scalar } \lambda, \text{ since } OA \parallel BC \\
 &= \underline{c} + \lambda \underline{a} \quad \checkmark
 \end{aligned}$$

Question 15(e)(ii)	Marks
· Provides correct solution.	3 marks
· Finds $\overrightarrow{OG}$ in terms of $\underline{a}$ and $\underline{c}$ using both $\overrightarrow{OE} + \overrightarrow{EG}$ and $\mu \overrightarrow{OB}$	2 marks
· Finds $\overrightarrow{OG}$ in terms of $\underline{a}$ and $\underline{c}$ using $\mu \overrightarrow{OB}$.	1 mark

Solution

The plan is to find $\overrightarrow{OG}$ in two different ways, then equate coefficients.
In order to achieve this, we will need to introduce two new scalars, μ and ν .

$$\begin{aligned}
 \overrightarrow{OG} &= \mu \overrightarrow{OB} \text{ for some scalar } \mu, \text{ since } OG \parallel OB \\
 &= \mu(\underline{c} + \lambda \underline{a}) \\
 &= \lambda \mu \underline{a} + \mu \underline{c} \quad \checkmark
 \end{aligned}$$

We also have

$$\begin{aligned}
 \overrightarrow{OG} &= \overrightarrow{OE} + \overrightarrow{EG} \\
 &= \overrightarrow{OE} + \nu \overrightarrow{EF} \text{ for some scalar } \nu, \text{ since } EG \parallel EF \\
 &= \frac{1}{3} \underline{a} + \nu (\overrightarrow{OF} - \overrightarrow{OE}) \text{ since } OE : EA = 1 : 2 \\
 &= \frac{1}{3} \underline{a} + \nu \left(\underline{c} + \frac{1}{2} \overrightarrow{CB} - \frac{1}{3} \underline{a} \right) \text{ since } F \text{ is the midpoint of } CB \\
 &= \frac{1}{3} \underline{a} + \nu \left(\underline{c} + \frac{\lambda}{2} \underline{a} - \frac{1}{3} \underline{a} \right) \text{ since } \overrightarrow{CB} = \lambda \overrightarrow{OA} \text{ from part (i)} \\
 &= \left(\frac{1}{3} + \frac{\lambda \nu}{2} - \frac{\nu}{3} \right) \underline{a} + \nu \underline{c} \quad \checkmark
 \end{aligned}$$

Question 15(e) (continued)

Equate the two expressions for $\overrightarrow{OG}$:

$$\lambda\mu\mathfrak{a} + \mu\mathfrak{c} = \left(\frac{1}{3} + \frac{\lambda\nu}{2} - \frac{\nu}{3}\right)\mathfrak{a} + \nu\mathfrak{c}$$

Equate coefficients of $\mathfrak{a}$ and $\mathfrak{c}$ (valid, since $\mathfrak{a}$ and $\mathfrak{c}$ are not parallel).

$$\lambda\mu = \frac{1}{3} + \frac{\lambda\nu}{2} - \frac{\nu}{3} \quad \textcircled{1}$$

$$\mu = \nu \quad \textcircled{2}$$

Substitute $\textcircled{2}$ into $\textcircled{1}$:

$$\lambda\mu = \frac{1}{3} + \frac{\lambda\mu}{2} - \frac{\mu}{3}$$

$$\frac{\lambda\mu}{2} = \frac{1 - \mu}{3}$$

$$3\lambda\mu = 2 - 2\mu$$

$$\mu(3\lambda + 2) = 2$$

$$\mu = \frac{2}{2 + 3\lambda}$$

Therefore, since $\overrightarrow{OG} = \lambda\mu\mathfrak{a} + \mu\mathfrak{c}$ from earlier, we have

$$\begin{aligned}\overrightarrow{OG} &= \lambda \left(\frac{2}{2 + 3\lambda} \right) \mathfrak{a} + \left(\frac{2}{2 + 3\lambda} \right) \mathfrak{c} \\ &= \left(\frac{2\lambda}{2 + 3\lambda} \right) \mathfrak{a} + \left(\frac{2}{2 + 3\lambda} \right) \mathfrak{c} \quad \checkmark\end{aligned}$$

Comments:

- Part (i) was very well done.
- Part (ii) was challenging.
- There was a common misconception that $\overrightarrow{OG}$ could be found by equating $\overrightarrow{EF}$ and $\overrightarrow{OB}$.
- The source of this misconception is that G is the intersection of the lines EF and OB .
- To make progress, students had to introduce two new scalars and solve simultaneously.

Question 16 (15 marks) Start on a NEW page

- (a) Suppose that a , b and c are the lengths of a right-angled triangle, where c is the hypotenuse.

Show that if $n \leq 2$, then $c^n \leq a^n + b^n$.

Since $a^2 + b^2 = c^2$

$c \geq a$ (A identity) ✓

$\therefore c^{n-2} \geq a^{n-2}$ as $n-2 \leq 0$

$a^2 c^{n-2} \geq a^n$ ①

Similarly, $b^2 c^{n-2} \leq b^n$ ② ✓

① + ② $\Rightarrow a^2 c^{n-2} + b^2 c^{n-2} \leq a^n + b^n$

$c^{n-2} (a^2 + b^2) \leq a^n + b^n$

$c^{n-2} \cdot c^2 \leq a^n + b^n$

$\therefore c^n \leq a^n + b^n$ ✓

Mark	Criteria
3	Correct solution.
2	Makes substantial progress in their proof by contradiction or setting up of inequality using the triangle inequality.
1	Correctly quotes the triangle inequality. OR Uses induction up to the inductive step but fails to go further.

Comments:

Poorly done. Students were awarded a mark if they successfully quoted the triangle inequality. Many students attempted induction, but this ultimately will not prove for all **real** numbers less than 2, i.e. $\{n \in \mathbb{R} \mid n \leq 2\}$. Additionally, students that proved by contradiction had to clearly show why the contradiction exists for all cases for n .

- (b) Show that the distance travelled by the particle before it has lost three-quarters of its initial speed is

$$\frac{15RU^2}{32gR - 15U^2} \text{ m,}$$

where R is the radius of the Earth and g is the acceleration at the Earth's surface.

$\ddot{x} = -\frac{k}{x^2}$

$v \frac{dv}{dx} = -\frac{k}{x^2}$

$\int_u^{1/4u} v dv = \int_R^x -\frac{k}{x^2} dx$

$\frac{1}{2} v^2 \Big|_u^{1/4u} = k \left[\frac{1}{x} \right]_R^x$

$\frac{1}{2} \left(\frac{1}{16} - 1 \right) u^2 = k \left(\frac{1}{x} - \frac{1}{R} \right)$

$-\frac{15}{32} u^2 = k \left(\frac{1}{x} - \frac{1}{R} \right)$

$x = \left(-\frac{15u^2}{32k} + \frac{1}{R} \right)^{-1}$ ✓

$x=R, \ddot{x} = -g \Rightarrow g = \frac{k}{R^2}$ ✓

$k = R^2 g$ ✓

$x = \left(\frac{-15u^2 + 32Rg}{32R^2g} \right)^{-1}$

$x = \frac{32R^2g}{32Rg - 15u^2}$ ✓

But $x = \text{distance from } O$

i.e. distance (d) travelled = $x - R$

$d = x - R$

$d = \frac{32R^2g - R(32Rg - 15u^2)}{32Rg - 15u^2}$

$\therefore d = \frac{15u^2 R}{32Rg - 15u^2}$ as required. ✓

Mark	Criteria
4	Correct solution.
3	Obtains $x = \frac{32R^2g}{32Rg - 15U^2}$ without error.
2	Obtains $x = \frac{32R^2g}{32Rg + 15U^2}$ by using $k = -R^2g$.
1	Makes some correct progress in obtaining x . OR Incorrectly obtains $x = \frac{32R^2g}{32Rg - 15U^2}$ from using $k = -R^2g$ OR Obtains $k = R^2g$

Comments:

Students made many errors from their signs. Students were commonly penalised for incorrectly obtaining $k = -R^2g$ and then using this expression to obtain $x = \frac{32R^2g}{32Rg - 15U^2}$ which cannot be done. **By forcing this expression, they were further penalised.** Additionally, at the end, many students did not realise they needed to subtract R from x as x is the distance from the centre of earth.

- (c) Let $I_n = \int_0^\alpha (\sec x + \tan x)^n dx$, where n is a non-negative integer and $0 < \alpha < \frac{\pi}{2}$.

(i) For $n \geq 1$, show that

$$\frac{1}{2}(I_{n+1} + I_{n-1}) = \frac{1}{n} ((\sec \alpha + \tan \alpha)^n - 1).$$

$$I_n = \int_0^\alpha (\sec x + \tan x)^{n-2} (\sec^2 x + \tan^2 x + 2 \sec x \tan x) dx$$

Note: $\frac{d}{dx}(\sec x + \tan x) = \sec x \tan x + \sec^2 x$
 $= \sec x (\tan x + \sec x)$
 $\sec^2 x = \tan^2 x + 1$
 $\tan^2 x = \sec^2 x - 1$

$$\begin{aligned} I_n &= \int_0^\alpha (\sec x + \tan x)^{n-2} \left(\underbrace{2 \sec^2 x + 2 \sec x \tan x}_{2 \sec x (\sec x + \tan x)} - 1 \right) dx \\ &= \int_0^\alpha \underbrace{2 \sec x (\sec x + \tan x)}_{f'(x) [f(x)]^{n-2} - \text{Rev. chain}} (\sec x + \tan x)^{n-2} dx - \underbrace{\int_0^\alpha (\sec x + \tan x)^{n-2} dx}_{I_{n-2}} \end{aligned}$$

$$I_n = 2 \left[\frac{(\sec x + \tan x)^{n-1}}{n-1} \right]_0^\alpha - I_{n-2}$$

$$= 2 \left(\frac{(\sec \alpha + \tan \alpha)^{n-1}}{n-1} - \left(\frac{1-0}{n-1} \right) \right) - I_{n-2}$$

$$I_n + I_{n-2} = \frac{2}{n-1} ((\sec \alpha + \tan \alpha)^{n-1} - 1) \quad [\div 2]$$

$$\frac{1}{2}(I_n + I_{n-2}) = \frac{1}{n-1} ((\sec \alpha + \tan \alpha)^{n-1} - 1)$$

Replace n by $n+1$,

$$\therefore \frac{1}{2}(I_{n+1} + I_{n-1}) = \frac{1}{n} ((\sec \alpha + \tan \alpha)^n - 1)$$

Mark	Criteria
3	Correct solution.
2	Makes substantial progress.
1	Uses trig identities to simplify the trig expression to $2 \sec^2 x + 2 \sec x \tan x - 1$

Comments:

Poorly done. Many students did not know what to differentiate and integrate when it came to setting up the integration by parts.

(ii) Deduce that

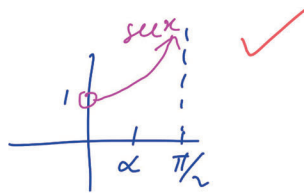
$$I_n < \frac{1}{n} \left((\sec \alpha + \tan \alpha)^n - 1 \right), \text{ for } n \geq 1.$$

$$\text{RTP: } I_n < \frac{1}{2} (I_{n+1} + I_{n-1})$$

$$\begin{aligned} \frac{1}{2} (I_{n+1} + I_{n-1}) &= \frac{1}{2} \int_0^\alpha (\sec x + \tan x)^{n+1} + (\sec x + \tan x)^{n-1} dx \\ &= \frac{1}{2} \int_0^\alpha (\sec x + \tan x)^{n-1} (\sec^2 x + \tan^2 x + 2 \sec x \tan x + 1) dx \\ &= \frac{1}{2} \int_0^\alpha (\sec x + \tan x)^{n-1} (2 \sec^2 x + 2 \sec x \tan x) dx \\ &= \frac{1}{2} \int_0^\alpha (\sec x + \tan x)^{n-1} \underbrace{2 \sec x (\sec x + \tan x)}_{\sec x (\sec x + \tan x) = \frac{d}{dx} (\sec x + \tan x)} dx \\ &= \int_0^\alpha (\sec x + \tan x)^n \sec x dx \quad \checkmark \end{aligned}$$

Since $\sec x > 1$ for $0 < x < \alpha$ where $0 < \alpha < \frac{\pi}{2}$

$$> \underbrace{\int_0^\alpha (\sec x + \tan x)^n dx}_{I_n}$$



$\therefore I_n < \frac{1}{2} (I_{n+1} + I_{n-1})$ as required.

Mark	Criteria
2	Correct solution.
1	Makes substantial progress by proving that the function is concave up in the domain $(0, \frac{\pi}{2})$.

Comments:

Poorly done. To obtain full marks students had to clearly prove that the $\sec x > 1$ in $x \in (0, \frac{\pi}{2})$.

- (d) Suppose that z_1, z_2 and $z_3 \in \mathbb{C}$ satisfy the following conditions simultaneously.

$$\begin{cases} |z_1| = |z_2| = |z_3| = 1 \\ \frac{z_1}{z_2} + \frac{z_2}{z_3} + \frac{z_3}{z_1} = 1 \quad (\star) \\ z_2 \neq z_3 \text{ and } z_3 \neq z_1. \end{cases}$$

Prove that if a, b and $c \in \mathbb{R}$, then $|az_1 + bz_2 + cz_3| = \sqrt{a^2 + b^2 + c^2 + 2ab}$.

let $z_1 = \text{cis } \alpha, z_2 = \text{cis } \beta, z_3 = \text{cis } \gamma$
as $|z_1| = |z_2| = |z_3| = 1$

$$\therefore \frac{\text{cis } \alpha}{\text{cis } \beta} + \frac{\text{cis } \beta}{\text{cis } \gamma} + \frac{\text{cis } \gamma}{\text{cis } \alpha} = 1 \quad \text{from } (\star)$$

$$\text{cis}(\alpha - \beta) + \text{cis}(\beta - \gamma) + \text{cis}(\gamma - \alpha) = 1$$

Equate the im. parts,

$$\sin(\alpha - \beta) + \sin(\beta - \gamma) + \sin(\gamma - \alpha) = 0$$

$$2 \sin\left(\frac{\alpha - \beta}{2}\right) \cos\left(\frac{\alpha - \beta}{2}\right) + 2 \sin\left(\frac{\beta - \gamma}{2}\right) \cos\left(\frac{\beta - \gamma}{2}\right) = 0$$

$$\sin(-\alpha) = -\sin \alpha$$

$$2 \sin\left(\frac{\alpha - \beta}{2}\right) \left[\cos\left(\frac{\alpha - \beta}{2}\right) - \cos\left(\frac{\beta - \gamma + \alpha}{2}\right) \right] = 0$$

$$\therefore \sin\left(\frac{\alpha - \beta}{2}\right) = 0 \quad \text{OR} \quad \cos\left(\frac{\alpha - \beta}{2}\right) = \cos\left(\frac{\beta - \gamma + \alpha}{2}\right)$$

$$\therefore \boxed{z_1 = z_2} \quad \checkmark$$

$$\text{as } \alpha = \beta$$

$$\alpha - \beta = \beta - \gamma + \alpha$$

$$\alpha = \gamma$$

$$\text{But } \alpha \neq \gamma \text{ as } \boxed{z_1 \neq z_3}$$

$$z_1 = z_2 = \text{cis } \alpha \Rightarrow 1 + \frac{z_1}{z_3} + \frac{z_3}{z_1} = 1$$

$$\frac{z_1}{z_3} + \frac{z_3}{z_1} = 0$$

$$z_1^2 + z_3^2 = 0$$

$$z_3^2 = -z_1^2 \Rightarrow z_3 = \pm i z_1 \quad \checkmark$$

Mark	Criteria
3	Correct solution.
2	Makes substantial progress or proves that $z_3 = \pm i z_1$.
1	Obtains $z_1 = z_2$

Comments:

Poorly done. Many students incorrectly used the vector formula $\vec{a} \cdot \vec{a} = |\vec{a}|^2$ to prove this result. However, this is invalid as z_1, z_2 and z_3 are complex numbers and have $i = \sqrt{-1}$ in their expression.

$$\star |z + w|^2 \neq (z + w) \cdot (z + w)$$

$$\begin{aligned}
 \therefore |az_1 + bz_2 + cz_3| &= |az_1 + bz_1 \pm icz_1| \\
 &= |z_1| |a+b \pm i| \\
 &= |z_1| \sqrt{(a+b)^2 + c^2} \\
 &= \sqrt{a^2 + b^2 + c^2 + 2ab} \quad \checkmark \\
 &\text{as required.}
 \end{aligned}$$

Alternate Solⁿ

$$\text{Let } \alpha = \frac{z_1}{z_2}, \quad \beta = \frac{z_2}{z_3}, \quad \gamma = \frac{z_3}{z_1}$$

$$\sum \alpha = 1 \quad (\text{given})$$

$$\alpha\beta\gamma = \frac{z_1}{z_2} \cdot \frac{z_2}{z_3} \cdot \frac{z_3}{z_1} = 1$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{z_1}{z_3} + \frac{z_2}{z_1} + \frac{z_3}{z_2}$$

$$\text{Now } |z_k| = 1 \quad \therefore |z_k|^2 = 1$$

$$\Rightarrow z_k \bar{z}_k = 1$$

$$\therefore \bar{z}_k = \frac{1}{z_k}$$

Sub back in!

$$\therefore \sum \alpha\beta = \frac{\bar{z}_3}{\bar{z}_1} + \frac{\bar{z}_1}{\bar{z}_2} + \frac{\bar{z}_2}{\bar{z}_3} = \left(\frac{\bar{z}_3}{z_1}\right) + \left(\frac{\bar{z}_1}{z_2}\right) + \left(\frac{\bar{z}_2}{z_3}\right)$$

$$= \frac{z_3}{z_1} + \frac{z_1}{z_2} + \frac{z_2}{z_3}$$

$$= 1 = 1$$

$\therefore \alpha, \beta, \gamma$ - roots of

$$x^3 - x^2 + x - 1 = 0$$

$$x^2(x-1) + (x-1) = 0$$

$$(x^2+1)(x-1) = 0 \Rightarrow x = \pm i, 1$$

Now since $z_2 \neq z_3$ and $z_3 \neq z_1$

$$\therefore \frac{z_2}{z_3} \neq 1, \frac{z_3}{z_1} \neq 1 \text{ i.e. } \alpha = 1 \Rightarrow \boxed{z_1 = z_2}$$

Now the rest follows!